Mathematical Image Modeling: Homework 6

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Jason Murphy 10:00

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Problem 1. Suppose X is a random variable with distribution function $\mu_X(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2}$. Prove that $\mathbb{E}(X) = 0$ and $\mathbb{E}(X^2) = 1$.

Solution. Since

$$\mu_X(t) = \frac{1}{\sqrt{2\pi}} e^{-t^2/2},,$$

this is the probability density function of the standard normal distribution. For a continuous random variable with density $f_X(t)$,

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} t f_X(t)\,\mathrm{d}t \quad \text{and} \quad \mathbb{E}(X^2) = \int_{-\infty}^{\infty} t^2 f_X(t)\,\mathrm{d}t..$$

For the first moment, we compute

$$\mathbb{E}(X) = \int_{-\infty}^{\infty} t \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\,\mathrm{d}t = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t e^{-t^2/2}\,\mathrm{d}t.$$

Observe that the function $te^{-t^2/2}$ is odd, since

$$(-t)e^{-(-t)^2/2} = -te^{-t^2/2}.$$

The integral of an odd function over a symmetric interval about 0 is 0, hence $\mathbb{E}(X) = 0$. For the second moment, we compute

$$\mathbb{E}(X^2) = \int_{-\infty}^{\infty} t^2 \frac{1}{\sqrt{2\pi}} e^{-t^2/2}\,\mathrm{d}t = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} t^2 e^{-t^2/2}\,\mathrm{d}t = \frac{\sqrt{2\pi}}{\sqrt{2\pi}} = 1.$$

Hence $\mathbb{E}(X) = 0$ and $\mathbb{E}(X^2) = 1$. □

Problem 2. Suppose X is a random variable with law $\mathrm{d}\mu_X = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} \delta_k$, where δ_k is a Dirac mass at k and $\lambda > 0$. Show that $\mathbb{E}(X) = \lambda$ and $\mathrm{Var}(X) = \lambda$. (*Hint:* For the variance, first compute $\mathbb{E}\{X(X-1)\}$.)

Solution. The law of X is

$$\mathrm{d}\mu_X = \sum_{k=0}^{\infty} \frac{\lambda^k}{k!} e^{-\lambda} \delta_k,$$

so

$$\mathbb{P}(X = k) = \frac{\lambda^k}{k!} e^{-\lambda},$$

for $k = 0, 1, 2, \ldots$. Thus X has the Poisson distribution with parameter λ. For the expectation, by definition,

$$\mathbb{E}(X) = \sum_{k=0}^{\infty} k\,\mathbb{P}(X = k) = \sum_{k=0}^{\infty} k \frac{\lambda^k}{k!} e^{-\lambda}.$$

The $k = 0$ term vanishes. Observe that

$$k \frac{\lambda^k}{k!} = \lambda \frac{\lambda^{k-1}}{(k-1)!}.$$

Hence

$$\mathbb{E}(X) = \lambda e^{-\lambda} \sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!}.$$

Let $j = k - 1$. Then

$$\sum_{k=1}^{\infty} \frac{\lambda^{k-1}}{(k-1)!} = \sum_{j=0}^{\infty} \frac{\lambda^j}{j!} = e^{\lambda}.$$

Therefore $\mathbb{E}(X) = \lambda$. For variance, we use the hint and compute $\mathbb{E}[X(X-1)]$. By definition,

$$\mathbb{E}[X(X-1)] = \sum_{k=0}^{\infty} k(k-1)\frac{\lambda^k}{k!}e^{-\lambda}.$$

The terms for $k = 0, 1$ vanish, so

$$\mathbb{E}[X(X-1)] = e^{-\lambda}\sum_{k=2}^{\infty} k(k-1)\frac{\lambda^k}{k!}.$$

Observe that

$$k(k-1)\frac{\lambda^k}{k!} = \lambda^2\frac{\lambda^{k-2}}{(k-2)!}.$$

Thus

$$\mathbb{E}[X(X-1)] = \lambda^2 e^{-\lambda}\sum_{k=2}^{\infty}\frac{\lambda^{k-2}}{(k-2)!}.$$

Let $j = k-2$. Then

$$\sum_{k=2}^{\infty}\frac{\lambda^{k-2}}{(k-2)!} = \sum_{j=0}^{\infty}\frac{\lambda^j}{j!} = e^{\lambda}.$$

Hence $\mathbb{E}[X(X-1)] = \lambda^2$. Now observe that

$$X^2 = X(X-1) + X.$$

Therefore

$$\mathbb{E}(X^2) = \mathbb{E}[X(X-1)] + \mathbb{E}(X) = \lambda^2 + \lambda.$$

Finally,

$$\operatorname{Var}(X) = \mathbb{E}(X^2) - (\mathbb{E}(X))^2 = (\lambda^2 + \lambda) - \lambda^2 = \lambda.$$

Thus $\mathbb{E}(X) = \lambda$ and $\operatorname{Var}(X) = \lambda$. □

Problem 3. Suppose we are trying to determine the intensity of some source in the presence of background noise. We therefore have two (independent) random variables S and N (the source and the noise), modeled as independent Poisson random variables with intensities λ_s and λ_n, respectively. By previous measurements, we have already determined λ_n. Our goal is to determine λ_s. Our sensor records only the total $X = S + N$.

(i) Show that X is a Poisson random variable with intensity $\lambda_s + \lambda_n$.

(ii) Show that if the sensor records N counts, then the maximum likelihood estimate for λ_s is $N - \lambda_n$. (Without loss of generality, you may assume $N > \lambda_n$.)

(To compute the maximum likelihood, you need to compute the probability that $X = N$ under the assumption that $\lambda_s = \lambda$, then optimize in the choice of λ.)

Solution to (i). Let S and N be independent Poisson random variables with parameters λ_s and λ_n, respectively. Thus

$$\mathbb{P}(S = k) = \frac{\lambda_s^k}{k!}e^{-\lambda_s} \quad \text{and} \quad \mathbb{P}(N = j) = \frac{\lambda_n^j}{j!}e^{-\lambda_n}.$$

Let $X = S + N$. For any integer $m \geq 0$,

$$\mathbb{P}(X = m) = \sum_{k=0}^{m}\mathbb{P}(S = k, N = m-k).$$

Since S and N are independent,

$$\mathbb{P}(S = k, N = m - k) = \mathbb{P}(S = k)\mathbb{P}(N = m - k).$$

Hence

$$\mathbb{P}(X = m) = \sum_{k=0}^{m} \frac{\lambda_s^k}{k!} e^{-\lambda_s} \frac{\lambda_n^{m-k}}{(m-k)!} e^{-\lambda_n}.$$

Factoring out the exponential term gives

$$\mathbb{P}(X = m) = e^{-(\lambda_s + \lambda_n)} \sum_{k=0}^{m} \frac{\lambda_s^k \lambda_n^{m-k}}{k!\,(m-k)!}.$$

Using

$$\binom{m}{k} = \frac{m!}{k!\,(m-k)!},$$

we rewrite the sum as

$$\sum_{k=0}^{m} \frac{\lambda_s^k \lambda_n^{m-k}}{k!\,(m-k)!} = \frac{1}{m!} \sum_{k=0}^{m} \binom{m}{k} \lambda_s^k \lambda_n^{m-k}.$$

Using the Binomial Theorem, we get

$$\mathbb{P}(X = m) = \frac{(\lambda_s + \lambda_n)^m}{m!} e^{-(\lambda_s + \lambda_n)}.$$

Thus X is a Poisson random variable with intensity $\lambda_s + \lambda_n$. □

Solution to (ii). Suppose the sensor records N counts, so $X = N$. From part (i),

$$X \sim \text{Poisson}(\lambda_s + \lambda_n).$$

To estimate λ_s, assume $\lambda_s = \lambda$ and compute the likelihood

$$L(\lambda) = \mathbb{P}(X = N \mid \lambda_s = \lambda).$$

Using the Poisson distribution,

$$L(\lambda) = \frac{(\lambda + \lambda_n)^N}{N!} e^{-(\lambda + \lambda_n)}.$$

To maximize this likelihood, it is convenient to maximize the log-likelihood:

$$\ell(\lambda) = \log L(\lambda) = N \log(\lambda + \lambda_n) - (\lambda + \lambda_n) - \log(N!\,).$$

Differentiating with respect to λ gives

$$\ell'(\lambda) = \frac{N}{\lambda + \lambda_n} - 1.$$

Setting this equal to zero yields

$$\frac{N}{\lambda + \lambda_n} = 1, \qquad \lambda + \lambda_n = N.$$

Thus $\lambda = N - \lambda_n$. Since we assume $N > \lambda_n$, this value is positive and therefore admissible. Hence the maximum likelihood estimate for λ_s is $\hat{\lambda}_s = N - \lambda_n$. □

Problem 4. Let $f : \mathbb{R}^n \to \mathbb{R}$ be a continuously differentiable function. Suppose that f has a unique global minimum at some point $x_* \in \mathbb{R}^n$. Let $x_0 \in \mathbb{R}^n$ be arbitrary and define $\{x_k\}$ recursively via

$$x_{k+1} = x_k - \lambda \nabla f(x_k),$$

where $\lambda > 0$.

(i) Show that if the sequence x_k converges, then it converges to x_*.

(ii) Fix $n = 1$. Suppose in addition that f is twice continuously differentiable and that $0 < m < f''(x) \leq M$ for some $m, M > 0$ and all $x \in \mathbb{R}$. Prove that the sequence x_k defined above converges provided λ is sufficiently small.

(This is the gradient descent algorithm with learning rate λ.)

Solution to (i). □

Solution to (ii). □

Problem 5. Let $f : \mathbb{R}^3 \to \mathbb{R}$ be infinitely differentiable and equal to zero outside of some ball. Show that

$$-\int_{\mathbb{R}^3} \frac{1}{|x|} \Delta f(x) \, \mathrm{d}x = 4\pi f(0),$$

where Δ is the Laplacian. This calculation shows that in three space dimensions, $-\Delta(1/|x|) = 4\pi\delta$ (where δ is the Dirac delta distribution).

Hint. Let $\varepsilon > 0$ and split the integral into the set $|x| \leq \varepsilon$ and the set $|x| > \varepsilon$. On the set $|x| > \varepsilon$, integrate by parts twice, using the identity

$$\int_S g \, \partial_j f \, \mathrm{d}x = -\int_S f \, \partial_j g \, \mathrm{d}x + \int_{\partial S} f g \nu_j \, \mathrm{d}A,$$

where ν is the outward normal vector on ∂S and dA is surface measure.

Solution. □

Problem 6. Recall that the Green's function $G = G(t, x, y)$ solves

$$\frac{1}{c^2(x)} \frac{\partial^2 G}{\partial t^2} - \Delta_x G = \delta(t)\delta(x - y),$$

with $G(t, x, y) = 0$ for $t < 0$. Show that

$$u(t, x) = \iint G(t - s, x, y) n(s, y) \, \mathrm{d}y \, \mathrm{d}s,$$

solves

$$\frac{1}{c^2(x)} \frac{\partial^2 u}{\partial t^2} - \Delta_x u = n(t, x).$$

Solution. Let

$$u(t, x) = \iint G(t - s, x, y) n(s, y) \, \mathrm{d}y \, \mathrm{d}s$$

Define the operator

$$L = \frac{1}{c^2(x)} \frac{\partial^2}{\partial t^2} - \Delta_x.$$

Since differentiation with respect to t and x can be passed under the integral, we obtain

$$Lu(t, x) = \iint \left(\frac{1}{c^2(x)} \frac{\partial^2}{\partial t^2} G(t - s, x, y) - \Delta_x G(t - s, x, y) \right) n(s, y) \, \mathrm{d}y \, \mathrm{d}s.$$

Because G satisfies

$$\frac{1}{c^2(x)} \frac{\partial^2 G}{\partial t^2} - \Delta_x G = \delta(t)\delta(x - y),$$

we substitute $t-s$ for t to obtain

$$\frac{1}{c^2(x)}\frac{\partial^2}{\partial t^2}G(t-s,x,y)-\Delta_x G(t-s,x,y)=\delta(t-s)\delta(x-y).$$

Hence

$$Lu(t,x)=\iint \delta(t-s)\delta(x-y)n(s,y)\,\mathrm{d}y\,\mathrm{d}s$$

Using the defining property of the Dirac delta,

$$\int \delta(t-s)f(s)\,\mathrm{d}s=f(t)\quad\text{and}\quad\int \delta(x-y)g(y)\,\mathrm{d}y=g(x),$$

the integrals collapse, $Lu(t,x)=n(t,x)$ Thus

$$\frac{1}{c^2(x)}\frac{\partial^2 u}{\partial t^2}-\Delta_x u=n(t,x),$$

so $u(t,x)$ solves the desired equation. $\square$